

Hints & Solutions

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Q1 (3) - Application of LHL and RHL

We have, $\lim_{x \rightarrow \frac{\pi}{2}^-} \left[\frac{x - \frac{\pi}{2}}{\cos x} \right]$

Now, $\lim_{x \rightarrow \frac{\pi}{2}^-} f(x) = \lim_{x \rightarrow \frac{\pi}{2}^-} \left[\frac{\frac{\pi}{2} - h - \frac{\pi}{2}}{\cos\left(\frac{\pi}{2} - h\right)} \right]$

$= \lim_{h \rightarrow 0} \left[\frac{-h}{\sin h} \right]$

$= -2$

and $\lim_{x \rightarrow \frac{\pi}{2}^+} f(x) = \lim_{x \rightarrow \frac{\pi}{2}^+} \left[\frac{\frac{\pi}{2} + h - \frac{\pi}{2}}{\cos\left(\frac{\pi}{2} + h\right)} \right]$

$= \lim_{h \rightarrow 0} \left[\frac{h}{-\sin h} \right]$

$= -2$

$\therefore \lim_{x \rightarrow 0} f(x) = -2$

Q2 (2) - Application of LHL and RHL

We have, $f(x) = \begin{cases} x^2 + 2, & x \geq 2 \\ 1 - x, & x < 2 \end{cases}$

and $g(x) = \begin{cases} 2x, & x > 1 \\ 3 - x, & x \leq 1 \end{cases}$

Now, $f[g(x)] = \begin{cases} [g(x)]^2 + 2, & g(x) \geq 2 \\ 1 - [g(x)], & g(x) < 2 \end{cases}$

$\Rightarrow f[g(x)] = \begin{cases} 4x^2 + 2, & x \geq 1 \\ (3 - x)^2 + 2, & x < 1 \end{cases}$

$\therefore \lim_{x \rightarrow 1} f[g(x)] = 6$

Q3 (2) - Application of LHL and RHL

We have, $\lim_{x \rightarrow a^-} \left(\frac{|x|^3}{a} - \left[\frac{x}{a} \right]^3 \right)$

$= \lim_{x \rightarrow a-h} \left(\frac{|a-h|^3}{a} - \left[\frac{a-h}{a} \right]^3 \right)$

$= \lim_{h \rightarrow 0} \left(\frac{|a|^3}{a} - \left[1 - \frac{h}{a} \right]^3 \right) \quad \left[\because \left[1 - \frac{h}{a} \right] = 0 \right]$

$= a^2$

Q4 (3) - Application of LHL and RHL

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We have, $\lim_{\theta \rightarrow 0} \left(\left[\frac{n \sin \theta}{\theta} \right] + \left[\frac{n \tan \theta}{\theta} \right] \right)$

We know that, $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$ $\left[\because \frac{\sin \theta}{\theta} < 1 \right]$

$\Rightarrow \lim_{\theta \rightarrow 0} \left[\frac{n \sin \theta}{\theta} \right] = n - 1$

Similarly, $\lim_{\theta \rightarrow 0} \left(\frac{n \tan \theta}{\theta} \right) = n$

$\left[\because \frac{\tan \theta}{\theta} > 1 \right]$

$\therefore \lim_{\theta \rightarrow 0} \left(\left[\frac{n \sin \theta}{\theta} \right] + \left[\frac{n \tan \theta}{\theta} \right] \right) = n - 1 + n = 2n - 1$

Q5 (2) - Application of LHL and RHL

We have, $(\sqrt[3]{1+a} - 1)x^2 + (\sqrt{1+a} - 1)x + \sqrt[6]{1+a} - 1 = 0$

Let $1 + a = y^6 \Rightarrow a \rightarrow 0, y \rightarrow 1$

$\therefore (y^2 - 1)x^2 + (y^3 - 1)x + y - 1 = 0$

$\Rightarrow (y - 1) \{ (y + 1)x^2 + (y^2 + y + 1)x + 1 \} = 0$

$\Rightarrow 2x^2 + 3x + 1 = 0$ $[\because y \rightarrow 1]$

$\Rightarrow (2x + 1)(x + 1) = 0$

$\Rightarrow x = -1, -\frac{1}{2}$

$\therefore \lim_{a \rightarrow 0^+} \alpha(a) = -1$

and $\lim_{a \rightarrow 0^+} \beta(a) = -\frac{1}{2}$

Q6 (3) - Series Expansion and Conversion to Definite Integrals

$\lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x - e^x)}{x^n}$

We know that, $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \infty$

and $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \infty$

$\Rightarrow \lim_{x \rightarrow 0} \frac{\left[\left\{ \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \infty \right) - 1 \right\} \right]}{\left[\left\{ \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \infty \right) - \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \infty \right) \right\} \right]} x^n$

$\Rightarrow \lim_{x \rightarrow 0} \frac{\left[\left(-\frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \infty \right) \right]}{\left[\left(-\frac{x}{1!} - \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \frac{x^5}{5!} + \frac{x^6}{6!} + \dots \infty \right) \right]} x^n$

$\Rightarrow \lim_{x \rightarrow 0} \frac{x^3 \left(-\frac{1}{2!} + \frac{x^2}{4!} - \frac{x^4}{6!} + \dots \infty \right) \left(-1 - x - \frac{x^2}{3!} + \frac{x^3}{4!} - \dots \infty \right)}{x^n}$

For the above limit to be finite and non-zero i.e. $\frac{1}{2}$

$\therefore n = 3$

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Q7 (3) - Series Expansion and Conversion to Definite Integrals

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e}{x} \\ = \lim_{x \rightarrow 0} \frac{e\left(1 - \frac{x}{2} + \frac{11x^2}{24} + \dots\right) - e}{x} \\ = \lim_{x \rightarrow 0} \frac{e \cdot x \left\{-\frac{1}{2} + \frac{11x}{24} + \dots\right\}}{x} = -\frac{e}{2} \end{aligned}$$

Q8 (2) - Series Expansion and Conversion to Definite Integrals

$$\lim_{n \rightarrow \infty} \frac{[x] + [2x] + \dots + [nx]}{n^2}$$

$$nx - 1 < [nx] \leq nx$$

Putting $n = 1, 2, 3, \dots, n$ and adding them

$$x \Sigma n - n < \Sigma [nx] \leq x \Sigma n$$

$$x \cdot \frac{\Sigma n}{n^2} - \frac{1}{n} < \frac{\Sigma [nx]}{n^2} \leq x \cdot \frac{\Sigma n}{n^2} \quad \dots (i)$$

$$\begin{aligned} \text{Now, } \lim_{n \rightarrow \infty} \left\{ x \cdot \frac{\Sigma n}{n^2} - \frac{1}{n} \right\} &= x \cdot \lim_{n \rightarrow \infty} \frac{\Sigma n}{n^2} - \lim_{n \rightarrow \infty} \frac{1}{n} \\ &= x \lim_{n \rightarrow \infty} \frac{n(n+1)}{2n^2} - 0 = \frac{x}{2} \end{aligned}$$

$$\text{and } \lim_{n \rightarrow \infty} x \cdot \frac{\Sigma n}{n^2} = x \cdot \lim_{n \rightarrow \infty} \frac{\Sigma n}{n^2} = \frac{x}{2}$$

As two limits are equal by Eq. (i) i.e. $x/2$.

Q9 (3) - Series Expansion and Conversion to Definite Integrals

$$\lim_{n \rightarrow \infty} \frac{[1^2x] + [2^2x] + \dots + [n^2x]}{n^3}$$

$$n^2x - 1 < [n^2x] \leq n^2x$$

Putting $n = 1, 2, 3, \dots, n$, we get

$$x \Sigma n^2 - n < \Sigma [n^2x] \leq x \Sigma n^2$$

$$x \frac{\Sigma n^2}{n^3} - \frac{1}{n^2} < \frac{\Sigma [n^2x]}{n^3} \leq x \frac{\Sigma n^2}{n^3} \dots (i)$$

$$\begin{aligned} \Rightarrow \lim_{n \rightarrow \infty} \left\{ x \cdot \frac{\Sigma n^2}{n^3} - \frac{1}{n^2} \right\} &= x \cdot \lim_{n \rightarrow \infty} \frac{\Sigma n^2}{n^3} - \lim_{n \rightarrow \infty} \frac{1}{n^2} \\ &= x \cdot \lim_{n \rightarrow \infty} \frac{n(n+1)(2n+1)}{6n^3} - 0 \end{aligned}$$

$$= x \cdot \lim_{n \rightarrow \infty} \frac{\left(1 + \frac{1}{n}\right)\left(2 + \frac{1}{n}\right)}{6}$$

$$= x \cdot \frac{2}{6} = \frac{x}{3}$$

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$$\text{and } \lim_{n \rightarrow \infty} x \frac{\Sigma n^2}{n^3} = \frac{x}{3}$$

As two limits are equal i.e. $x/3$.

Q10 (4) - Series Expansion and Conversion to Definite Integrals

$$\begin{aligned} \lim_{n \rightarrow \infty} \sum_{i=0}^n \frac{1}{\sqrt{n^2 + r^2}} \\ = \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \sum_{r=0}^n \frac{1}{\sqrt{1 + (r/n)^2}} \\ = \int_0^1 \frac{dx}{\sqrt{1+x^2}} \Rightarrow \left[\log \left| x + \sqrt{1+x^2} \right| \right]_0^1 \\ = \log(1 + \sqrt{2}) \end{aligned}$$

Q11 (3) - Series Expansion and Conversion to Definite Integrals

$$\text{Given, } f'(0) = \lim_{n \rightarrow \infty} n \left[f\left(\frac{1}{n}\right) \right] \text{ and } f(0) = 0$$

$$\begin{aligned} \therefore \lim_{n \rightarrow \infty} \left[\frac{2}{\pi} (n+1) \cos^{-1}\left(\frac{1}{n}\right) - n \right] \\ = \lim_{n \rightarrow \infty} n \left[\frac{2}{\pi} \left(1 + \frac{1}{n}\right) \cos^{-1} \frac{1}{n} - 1 \right] \end{aligned}$$

$$\text{Let } f(x) = \frac{2}{\pi} (1+x) \cos^{-1} x - 1$$

$$\begin{aligned} \text{Now, } f'(x) &= \frac{2}{\pi} \left[(1+x) \frac{-1}{\sqrt{1-x^2}} + \cos^{-1} x \right] \\ f'(0) &= \frac{2}{\pi} \left[-1 + \frac{\pi}{2} \right] = \frac{2}{\pi} \left[\frac{\pi-2}{2} \right] = \left(\frac{\pi-2}{\pi} \right) \end{aligned}$$

Q12 (2) - Continuity

$$\text{We have, } f(x) = |[x] - 1|$$

Let a be any integer.

$$\therefore \lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a+h} |[a+h] - 1| = |a-1| \quad [\because [a+h] = a]$$

$$\text{and } \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a-h} |[a-h] - 1| = |a-2| \quad [\because [a-h] = a-1]$$

$$\therefore \lim_{x \rightarrow a^+} f(x) \neq \lim_{x \rightarrow a^-} f(x)$$

So, $f(x)$ is discontinuous at integer point.

$\therefore f(x)$ is continuous, $\forall x \in \mathbb{R} - \text{Integer}$.

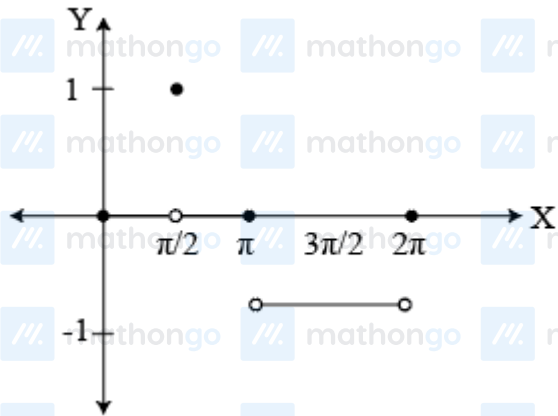
Q13 (2) - Continuity

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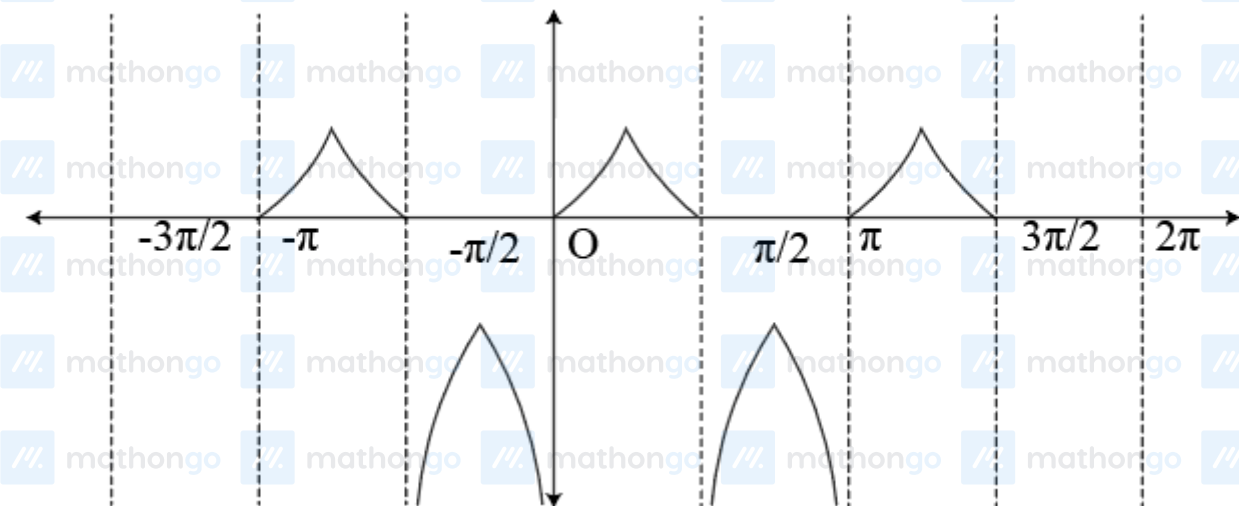
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Graph of $y = [\sin x]$ is $x \in [0, 2\pi]$



From the graph, $f(x)$ is discontinuous at $x = \frac{\pi}{2}, \pi, 2\pi$.

Q14 (2) - Continuity



From the graph, $f(x)$ is discontinuous at

$$x = \pm\pi, \pm\pi/2, 0, \dots$$

$$\therefore x = \frac{n\pi}{2}$$

Graph of $\min\{\tan x, \cot x\}$ is

Q15 (3) - Continuity

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$$\text{We have, } f(x) = \begin{cases} x + 2, & x < 1 \\ 4x - 1, & 1 \leq x \leq 3 \\ x^2 + 5, & x > 3 \end{cases}$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} x + 2 = 1 + 2 = 3 \dots \text{(i)}$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} 4x - 1 = 4 - 1 = 3 \dots \text{(ii)}$$

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} 4x - 1 = 12 - 1 = 11 \dots \text{(iii)}$$

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} x^2 + 5 = 9 + 5 = 14 \dots \text{(iv)}$$

$$\text{From Eqs. (i) and (ii), } \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = f(1) = 3$$

$$\therefore f(x) \text{ is continuous at } x = 1$$

$$\text{From Eqs. (iii) and (iv), } \lim_{x \rightarrow 3^-} f(x) \neq \lim_{x \rightarrow 3^+} f(x)$$

$$\therefore f(x) \text{ is discontinuous at } x = 3$$

Q16 (1) - Continuity

$$\text{We have, } f(x) = \begin{cases} [x] + [-x], & x \neq 2 \\ \lambda, & x = 2 \end{cases}$$

$$f(x) \text{ is continuous at } x = 2$$

$$\therefore \lim_{x \rightarrow 2} f(x) = f(2)$$

$$\Rightarrow \lim_{x \rightarrow 2} [x] + [-x] = \lambda \quad [\because f(2) = \lambda]$$

$$\Rightarrow -1 = \lambda$$

$$\left[\begin{array}{l} \because [x] + [-x] = 0, x \in \text{Integer} \\ \text{and } [x] + [-x] = -1, x \notin \text{Integer} \end{array} \right]$$

$$\therefore \lambda = -1$$

Q17 (4) - Continuity

$$\text{We have, } f(x) = \frac{x - e^x + \cos 2x}{x^2}, x \neq 0$$

$$f(x) \text{ is continuous at } x = 0 \therefore \lim_{x \rightarrow 0} f(x) = f(0)$$

$$\Rightarrow f(0) = \lim_{x \rightarrow 0} \frac{x - e^x + \cos 2x}{x^2}$$

$$\Rightarrow f(0) = \lim_{x \rightarrow 0} \frac{1 - e^x + 2 \sin 2x}{2x} \quad [\text{applying L' Hospital's rule}]$$

$$\Rightarrow f(0) = \lim_{x \rightarrow 0} \frac{-e^x - 4 \cos 2x}{2}$$

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$$\Rightarrow f(0) = \frac{-5}{2}$$

$$\therefore [f(0)] = \left[\frac{-5}{2} \right] = -3$$

$$\text{and } \{f(0)\} = \left\{ \frac{-5}{2} \right\} = 0.5$$

$$\therefore [f(0)]\{f(0)\} = -3 \times 0.5 = -1.5$$

Q18 (4) - Continuity

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{2}{e^{2x}-1} \right) = \lim_{x \rightarrow 0} \frac{e^{2x}-1-2x}{x(e^{2x}-1)}$$

$$= \lim_{x \rightarrow 0} \frac{2e^{2x}-2}{(e^{2x}-1)+2xe^{2x}} \text{ [by L' Hospital's rule]}$$

$$= \lim_{x \rightarrow 0} \frac{4e^{2x}}{4e^{2x}+4xe^{2x}} \text{ [by L' Hospital's rule]}$$

$$= 1$$

So, $f(x)$ is continuous at $x = 0$.

$$\therefore \lim_{x \rightarrow 0} f(x) = f(0)$$

$$\Rightarrow f(0) = 1$$

Q19 (4) - Continuity

$$\text{Given, } f(x) = [x^2 + 1] = [x^2] + 1$$

Now, x^2 is monotonic in the range of $[1, 3]$.

Hence, $[x^2]$ is discontinuous when x^2 is integer.

$$\therefore x^2 = 2, 3, 4, \dots, 9$$

$$\text{or } x = \sqrt{2}, \sqrt{3}, \sqrt{4}, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{8}, \sqrt{9}$$

Note $f(x)$ is right continuous at $x = 1$ but not left continuous at $x = 3$.

$$\therefore \lim_{x \rightarrow 1^+} [x^2 + 1] = 2 = f(1)$$

$$\text{and } \lim_{x \rightarrow 3^-} [x^2 + 1] = 9 \neq 10 = f(3)$$

Hence, $f(x)$ is continuous for all x except eight points.

Q20 (2) - Continuity

$$f(x) = [2x^3 - 5] \text{ in } [1, 2) \text{ is discontinuous and } [.] \text{ is G.I.F.}$$

We know that x^3 is continuous $\Rightarrow 2x^3 - 5$ is also continuous.

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Now, range of $2x^3 - 5$ for $x \in [1, 2)$ will be

$$-3 \leq 2x^3 - 5 < 11 \\ \Rightarrow (2x^3 - 5) \in [-3, 11)$$

We know G.I.F is discontinuous at integers, $f(x)$ is also a G.I.F. with domain as $[-3, 11)$, so $f(x)$ will be discontinuous at integer in $[-3, 11)$

$\Rightarrow$ No. of point of discontinuity = no. of integers in $[-3, 11) = 14$

Q21 (2) - Continuity

$$\text{Given, } f(x) = \begin{cases} |x+1|, & \text{if } x < -2 \\ 2x+3, & \text{if } -2 \leq x < 0 \\ x^2+3, & \text{if } 0 \leq x < 3 \\ x^3-15, & \text{if } 3 \leq x \end{cases} \Rightarrow f(x) = \begin{cases} -(x+1), & \text{if } x < -2 \\ 2x+3, & \text{if } -2 \leq x < 0 \\ x^2+3, & \text{if } 0 \leq x < 3 \\ x^3-15, & \text{if } 3 \leq x \end{cases}$$

At $x = -2$

$$f(-2^-) = \lim_{h \rightarrow 0} -(-2 - h + 1) = 1$$

$$f(-2^+) = \lim_{h \rightarrow 0} 2(-2 + h) + 3 = -1$$

$f(x)$ is discontinuous at $x = -2$. Now, at $x = 0$

$$f(0^-) = f(0^+) = f(0) = 3$$

$\therefore f(x)$ is continuous at $x = 0$ At $x = 3$

$$f(3^-) = f(3^+) = f(3) = 12$$

$f(x)$ is continuous at $x = 3$.

Hence, only point of discontinuity is $x = -2$.

Q22 (3) - Continuity

As, $f(x)$ is continuous for all $x \in R$ and $f(x) = 5, \forall x \in \text{irrational number}$

$\Rightarrow f(x)$ must be constant.

$$\therefore f(x) = f(3) = 5$$

Q23 (1) - Continuity

$$\text{Given, } f(x^2 + 1) = \frac{2}{f(2^x) - 1}, f(x) > 0 \dots (i)$$

$$\text{Put } t = x^2 + 1, \text{ then } x = \sqrt{t - 1}$$

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$$\therefore f(t) = \frac{2}{f(2^{\sqrt{t-1}})-1}$$

$$f(x) = \frac{2}{f(2^{\sqrt{x-1}})-1} \dots (ii)$$

$$\text{From Eq. (i), } f(1) = \frac{2}{f(1)-1} \Rightarrow f(1)^2 - f(1) = 2$$

$$\Rightarrow f(1)^2 - f(1) - 2 = 0 \Rightarrow f(1) = 2, -1$$

$$\Rightarrow f(1) = 2 [\because f(x) > 0 \text{ for all } x \in \mathbb{R}]$$

$$\text{Now, } \lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \frac{2}{f(2^{\sqrt{x-1}})-1} \text{ [using Eq. (ii)]}$$

$$= \frac{2}{f(1)-1} = \frac{2}{2-1}$$

$$= 2$$

Q24 (2) - Continuity

$$\text{We have, } h(x) = \lim_{n \rightarrow \infty} \frac{x^{n+1}f(x) + x^2}{x^n + g(x)}, x > 0$$

$$h(x) = \begin{cases} \frac{x^2}{g(x)}, & 0 < x < 1 \\ f(x), & x > 1 \\ \frac{f(1)+1}{1+g(1)} & x = 1 \end{cases}$$

Since, $h(x)$ is continuous for $x > 1$

$$\therefore h(1) = h(1^+) = h(1^-)$$

$$\Rightarrow \frac{1}{g(1)} = f(1)$$

$$\Rightarrow f(1) \cdot g(1) = 1$$

Q25 (1) - Differentiability

$$\text{Given } f(x) = x^3 \text{ Sgn}(x)$$

$$\text{or } f(x) = \begin{cases} x^3, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ -x^3, & \text{if } x < 0 \end{cases}$$

To check differentiability at $x = 0$

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$$f'(0^-) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{-(0-h)^3 - 0}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{h^3}{-h}$$

$$= 0$$

$$f'(0^+) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(0+h)^3 - 0}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h^3}{h} = 0$$

$\therefore f(x)$ is differentiable at $x = 0$

Q26 (4) - Differentiability

$$f(x) = (x^2 - 1)|x^2 - 3x + 2| + \cos(|x|)$$

$$= (x-1)(x+1) \cdot |(x-1)(x-2)| + \cos x$$

$(x-1)|x-1|$ and $\cos x$ are differentiable for all x , but $|x-2|$ is not differentiable at $x = 2$.

Hence, $f(x)$ is non-differentiable at $x = 2$.

Q27 (3) - Differentiability

$$\text{Given, } f(x) = \begin{cases} xe^{-\left(\frac{1}{|x|} + \frac{1}{x}\right)}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} x \cdot e^{-\left(-\frac{1}{x} + \frac{1}{x}\right)} = x, & x < 0 \\ x \cdot e^{-2/x}, & x > 0 \\ 0, & x = 0 \end{cases}$$

$$\text{Clearly, } \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0} x = 0$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0} x \cdot e^{-2/x} = 0$$

$$\text{Also, } f(0) = 0$$

$$\therefore \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$$

So, $f(x)$ is continuous at $x = 0$.

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$$\text{Now, } Lf'(0^-) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{0 - h - 0}{-h} = 1$$

$$Rf'(0^+) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$= \frac{(0+h) \cdot e^{-\frac{2}{0+h}} - 0}{h}$$

$$= 0$$

$$\therefore Lf'(0^-) \neq Rf'(0^+)$$

$$\therefore f(x) \text{ is not differentiable at } x = 0$$

Q28 (1) - Differentiability

We have,

$$f(x) = [x] \sin \pi x$$

$$\therefore Lf'(k^-) = \lim_{x \rightarrow k^-} \frac{f(x) - f(k)}{x - k}$$

$$\lim_{h \rightarrow 0} \frac{f(k-h) - f(k)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{[k-h] \sin(\pi k - \pi h) - [k] \sin \pi k}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{(-1)^{k-1} [k-h] \sin \pi h - 0}{-h}$$

$$= \lim_{h \rightarrow 0} (-1)^k \frac{[k-h] \sin \pi h}{h}$$

$$= (-1)^k (k-1) \lim_{h \rightarrow 0} \frac{\sin \pi h}{h}$$

$$= (-1)^k (k-1) \pi$$

Q29 (1) - Differentiability

$$\text{Given, } f(x) = ||x| - 1|$$

It is not differentiable, when $|x| = 0$ i.e. $x = 0$. and when $|x| - 1 = 0$ i.e. $x = \pm 1$

Q30 (3) - Differentiability

RHL at $x = 0$

$$= \lim_{h \rightarrow 0} h + \{h\} + h \sin\{h\}$$

$$= \lim_{h \rightarrow 0} h + h + h \sin h = 0$$

LHL at $x = 0$

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$$\begin{aligned}
 &= \lim_{h \rightarrow 0} (-h) + \{-h\} - h \cdot \sin\{-h\} \\
 &= \lim_{h \rightarrow 0} -h + 1 - h - h \cdot \sin(1 - h) = 1
 \end{aligned}$$

Since, RHL $\neq$ LHL

$\therefore$ Neither continuous nor differentiable at $x = 0$

Q31 (3) - Differentiability

We have, $f(x) = x^3$ and $g(x) = |x|$

$$\therefore \text{gof}(x) = |f(x)| = |x^3|$$

$$\text{and fog}(x) = [g(x)]^3 = |x|^3$$

Clearly, at $x = 0$, both are differentiable i.e. derivable.

Q32 (2) - Differentiability

(1) Let take an example. $f(x) = \sin x$ is differentiable at $x = \pi$, but $|\sin x|$ is not differentiable at $x = \pi$

Hence, option (a) is incorrect.

(2) True, because $f(x) = \sin x$ is continuous at $x = a$, then $f(x) = |\sin x|$ is also continuous at $x = a$.

Q33 (4) - Differentiability

We have,

$$f(x) = x[x] = \begin{cases} -x & , \text{ for } -1 \leq x < 0 \\ 0 & , \text{ for } 0 \leq x < 1 \\ x & , \text{ for } 1 \leq x < 2 \\ 2x & , \text{ for } 2 \leq x < 3 \\ 3x & , \text{ for } x = 3 \end{cases}$$

which is discontinuous at $x = 0, 1, 2$ and 3 .

$$f'(x) = \begin{cases} -1 & , \text{ for } -1 \leq x < 0 \\ 0 & , \text{ for } 0 \leq x < 1 \\ 1 & , \text{ for } 1 \leq x < 2 \\ 2 & , \text{ for } 2 \leq x < 3 \\ 3 & , \text{ for } x = 3 \end{cases}$$

Clearly, $f(x)$ is not differentiable at $x = 0, 1, 2$ and 3 .

Q34 (3) - Differentiability

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Hints & Solutions

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At $x = 0$

$$\begin{aligned}
 Lf'(0) &= \frac{f(0-h) - f(0)}{-h} \\
 &= \lim_{h \rightarrow 0} \frac{-h \sin\left(-\frac{1}{h}\right) - 0}{-h} \\
 &= \lim_{h \rightarrow 0} -\sin\left(\frac{1}{h}\right)
 \end{aligned}$$

which is not unique.

Hence, $f(x)$ is not differentiable at $x = 0$.

$$\text{Now, } Lg'(0) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h}$$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \frac{n^2 \sin\left(-\frac{1}{h}\right) - 0}{-h} \\
 &= \lim_{h \rightarrow 0} h \sin\left(\frac{1}{h}\right) = 0 \times (\text{Finite}) = 0
 \end{aligned}$$

$$\text{and } Rg'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h^2 \sin\left(\frac{1}{h}\right) - 0}{h}$$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} h \sin \frac{1}{h} \\
 &= 0 \times (\text{Finite}) = 0
 \end{aligned}$$

 $\therefore g(x)$ is differentiable at $x = 0$ and $h(x)$ is also differentiable at $x = 0$

Q35 (4) - Differentiability

We have,

$$f(x) = \begin{cases} \frac{1}{2}(-x-1), & \text{if } x < -1 \\ \tan^{-1} x, & \text{if } -1 \leq x \leq 1 \\ \frac{1}{2}(x-1), & \text{if } x > 1 \end{cases}$$

 $f(x)$ is discontinuous at $x = -1$ and $x = 1$ $\therefore$ Domain of $f'(x)$ is $R - \{-1, 1\}$.